

Wazuh SIEM Homelab: Single-Node Detection Lab

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1 Goal and Architecture

The goal of this lab was to build a working SIEM detection pipeline and prove it end to end: generate an attack, collect the logs, fire a stock rule, then write and validate a custom detection. The stack is the official Wazuh single-node Docker deployment (version 4.14.8): one manager, one indexer, one dashboard, plus a Dockerized Ubuntu victim with an enrolled Wazuh agent. The detection scenario covers SSH password guessing, mapped to MITRE T1110.001 (Brute Force: Password Guessing).

```
attacker (host) --> victim-ubuntu (sshd, agent)
victim-ubuntu --auth.log--> manager:1514
manager --filebeat--> indexer:9200 --> dashboard:443
```

2 Build: Single-Node Stack

I followed the official single-node Docker guide: cloned `wazuh-docker`, generated indexer certificates with the `wazuh-certs-generator` container, and started the stack with `docker compose up -d`. The host (Arch Linux, 8 GB RAM) already exceeded the indexer memory-map requirement, and the required ports (443, 9200, 1514, 1515) were free. First boot took about two minutes while the indexer built its indexes; cluster health reported **green** with one node.

```
[null@arch single-node]$ docker compose up -d
[+] up 59/59
✓ Image wazuh/wazuh-dashboard:4.14.8      Pulled      501.8s
✓ Image wazuh/wazuh-manager:4.14.8        Pulled      263.2s
✓ Image wazuh/wazuh-indexer:4.14.8        Pulled      460.2s
✓ Volume single-node_filebeat_etc         Created     0.0s
✓ Volume single-node_wazuh_active_response Created     0.0s
✓ Volume single-node_filebeat_var          Created     0.0s
✓ Volume single-node_wazuh_integrations    Created     0.0s
✓ Volume single-node_wazuh-indexer-data    Created     0.0s
✓ Volume single-node_wazuh-dashboard-custom Created     0.0s
✓ Volume single-node_wazuh-dashboard-config Created     0.0s
✓ Volume single-node_wazuh_api_configuration Created     0.0s
✓ Volume single-node_wazuh_agentless       Created     0.0s
✓ Volume single-node_wazuh_logs            Created     0.0s
✓ Volume single-node_wazuh_wodles          Created     0.0s
✓ Volume single-node_wazuh_queue           Created     0.0s
✓ Volume single-node_wazuh_etc             Created     0.0s
✓ Volume single-node_wazuh_var_multigroups Created     0.0s
✓ Container single-node-wazuh.indexer-1    Started     0.8s
✓ Container single-node-wazuh.manager-1    Started     0.6s
✓ Container single-node-wazuh.dashboard-1  Started     0.9s
[null@arch single-node]$
```

Figure 1: Stack boot: images pulled, volumes created, all three containers started

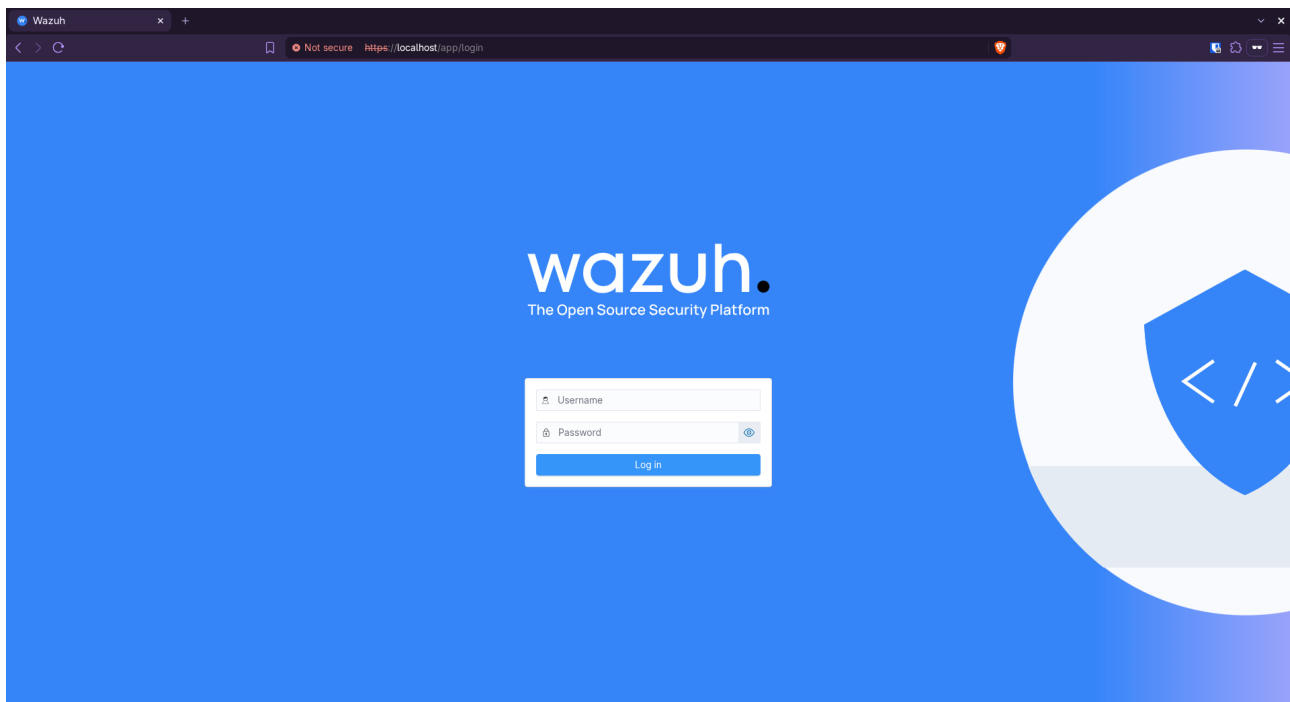


Figure 2: Dashboard reachable at `https://localhost`

3 Victim Enrollment

The victim is an `ubuntu:22.04` container named `victim-ubuntu`, attached to the compose network `single-node_default` so the manager hostname `wazuh.manager` resolves. Inside it I installed the Wazuh agent (4.14.8, matching the manager) and OpenSSH server. Two issues came up: the agent ships with the placeholder `MANAGER_IP` in `ossec.conf`, which I replaced with `wazuh.manager`, and enrollment is passwordless on this manager (`use_password=no`), so `agent-auth -m wazuh.manager -A victim-ubuntu` was enough. The manager then listed the agent as ID 001, Active.

```
[null@arch ~]$ docker run -d --name victim-ubuntu --hostname victim-ubuntu --network single-node_default ubuntu:22.04 s
leep infinity
Unable to find image 'ubuntu:22.04' locally
22.04: Pulling from library/ubuntu
20c3783cc497: Pull complete
Digest: sha256:b8b6ee6aa931ecd9d0d952abc34dc0e5f7c6a30c6bb71b079fe399fde0329c02
Status: Downloaded newer image for ubuntu:22.04
7b1aa58e8d0fee4c81e320cebfa7ffe3955ea0c4e159ed11730d78d4fe4692d
[null@arch ~]$ docker ps --filter name=victim-ubuntu
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
7b1aa58e8d0f	ubuntu:22.04	"sleep infinity"	9 seconds ago	Up 7 seconds		victim-ubuntu

Figure 3: Victim container launch: image pull and running status

```
root@victim-ubuntu:/# curl -s https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.8-1_amd64.deb
-o /tmp/agent.deb
root@victim-ubuntu:/# dpkg -i /tmp/agent.deb || apt-get install -f -y
Selecting previously unselected package wazuh-agent.
(Reading database ... 10528 files and directories currently installed.)
Preparing to unpack /tmp/agent.deb ...
Unpacking wazuh-agent (4.14.8-1) ...
Setting up wazuh-agent (4.14.8-1) ...
root@victim-ubuntu:/# agent-auth
bash: agent-auth: command not found
root@victim-ubuntu:/# /var/ossec/bin/agent-auth -m wazuh.manager -A victim-ubuntu
2026/09/24 09:57:08 agent-auth: INFO: Started (pid: 5096).
2026/09/24 09:57:08 agent-auth: INFO: Requesting a key from server: wazuh.manager
2026/09/24 09:57:08 agent-auth: INFO: No authentication password provided
2026/09/24 09:57:08 agent-auth: INFO: Using agent name as: victim-ubuntu
2026/09/24 09:57:08 agent-auth: INFO: Waiting for server reply
2026/09/24 09:57:08 agent-auth: INFO: Valid key received
```

Figure 4: Agent package install and passwordless enrollment (valid key received)

```

root@victim-ubuntu:/# service wazuh-agent start
Starting Wazuh v4.14.8...
Started wazuh-execd...
Started wazuh-agentd...
Started wazuh-syscheckd...
Started wazuh-logcollector...
Started wazuh-modulesd...
Completed.
root@victim-ubuntu:/# service ssh start
* Starting OpenBSD Secure Shell server sshd
root@victim-ubuntu:/# |

```

Figure 5: Agent and SSH services started on the victim

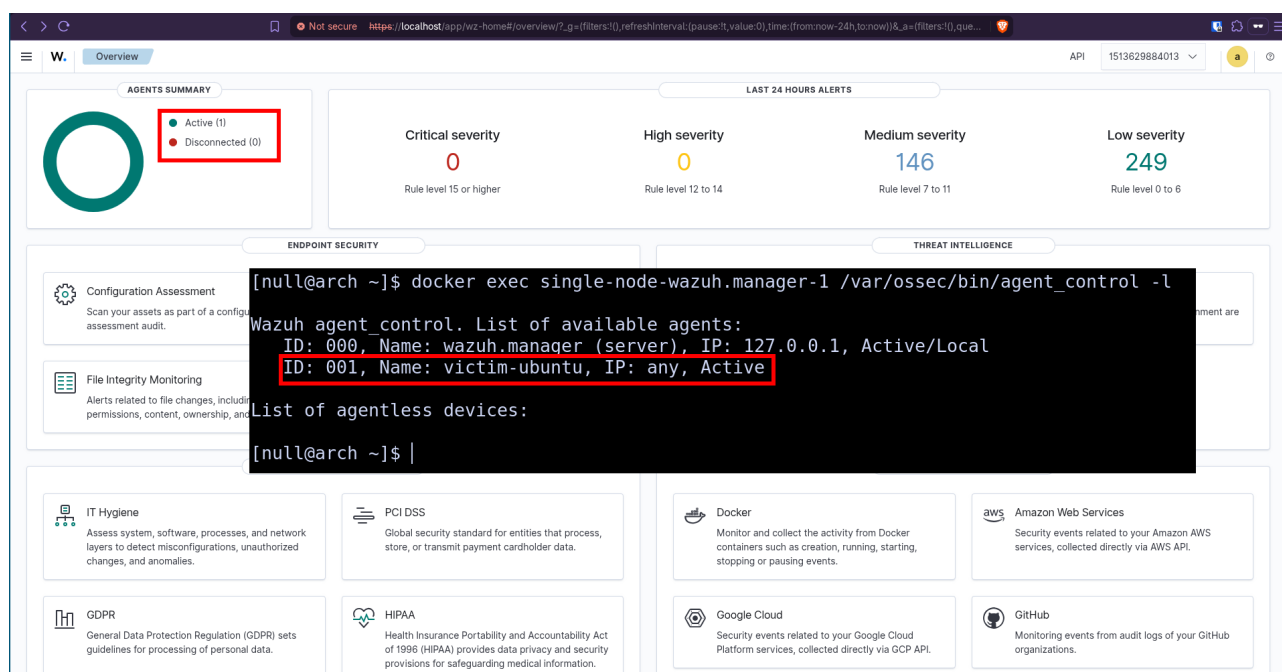


Figure 6: Agent 001 Active: dashboard overview (1 active, alert counters) and manager CLI

4 Problem: Missing Auth Logs

The first brute force produced no alerts. Debugging showed the attack did reach the victim (orphaned `sshd` processes at the attack time) but was logged nowhere: the container ships without a `syslog` daemon, so `/var/log/auth.log` did not exist, and the agent's `journald` source reads nothing without `systemd`. The fix was to install `rsyslog` and run `rsyslogd` directly (no `init` system in the container, so `service rsyslog start` didn't work), then add a file monitor to the agent configuration (Figure 7):

```

<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/auth.log</location>
</localfile>

```

After restarting the agent, failed logins appeared in `auth.log` and flowed to the manager.

```
GNU nano 6.2 /var/ossec/etc/ossec.conf
<localfile>
  <log_format>full_command</log_format>
  <command>netstat -tulpn | sed 's/\([[:alnum:]]\+\)\ \([[:digit:]]\+\) \+
  <alias>netstat listening ports</alias>
  <frequency>360</frequency>
</localfile>

<localfile>
  <log_format>full_command</log_format>
  <command>last -n 20</command>
  <frequency>360</frequency>
</localfile>

<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/auth.log</location>
</localfile>

<!-- Active response -->
<active-response>
  <disabled>no</disabled>
  <ca_store>etc/wpk_root.pem</ca_store>
  <ca_verification>yes</ca_verification>
</active-response>
```

Figure 7: `auth.log` file monitor added to the agent `ossec.conf`

5 Detection 1: Stock Rule (Password Guessing)

The attack is a minimal password loop from the host against the victim's IP:

```
for i in 1 2 3 4 5 6 7 8; do
  sshpass -p x ssh -o StrictHostKeyChecking=no fakeuser@VICTIM_IP exit
done
```

Each attempt logs `Failed password for invalid user`, which fires stock rule **5710** (level 5, “Attempt to login using a non-existent user”) rather than the generic 5716, because the message names an invalid user. PAM failures fire **5503**, and bursts also trip the stock aggregate **5551** (level 10, multiple failed logins). A repeat run confirmed the alerts land in Threat Hunting within about a minute.

```
[null@arch ~]$ docker inspect victim-ubuntu --format '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}'
172.21.0.5
[null@arch ~]$
[null@arch ~]$ ping -c 1 172.21.0.5
PING 172.21.0.5 (172.21.0.5) 56(84) bytes of data.
64 bytes from 172.21.0.5: icmp_seq=1 ttl=64 time=0.048 ms

--- 172.21.0.5 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.048/0.048/0.048/0.000 ms
[null@arch ~]$
[null@arch ~]$ for i in $(seq 1 10); do sshpass -p x ssh -o StrictHostKeyChecking=no fakeuser@172.21.0.5 exit; done;
Warning: Permanently added '172.21.0.5' (ED25519) to the list of known hosts.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
[null@arch ~]$ |
```

Figure 8: First brute-force loop and resulting auth.log traces on the victim

457 hits				
Sep 23, 2026 @ 13:56:16.799 - Sep 24, 2026 @ 13:56:16.800				
Export Formatted	Reset view	637 available fields	Columns	Density
1 fields sorted	Full screen			
timestamp	agent.name	rule.description	rule.level	rule.id
Sep 24, 2026 @ 13:51:05.9...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 13:51:04.0...	victim-ubuntu	PAM: User login failed.	5	5503
Sep 24, 2026 @ 13:51:03.9...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 13:51:01.9...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 13:51:00.0...	victim-ubuntu	PAM: User login failed.	5	5503
Sep 24, 2026 @ 13:50:59.9...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 13:50:57.9...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 13:50:56.0...	victim-ubuntu	PAM: Multiple failed logins in a small period of time.	10	5551
Sep 24, 2026 @ 13:50:55.9...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 13:50:53.9...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 13:50:52.0...	victim-ubuntu	PAM: User login failed.	5	5503
Sep 24, 2026 @ 13:50:51.9...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 13:50:49.9...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 13:50:48.0...	victim-ubuntu	PAM: User login failed.	5	5503
Sep 24, 2026 @ 13:50:47.9...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710

Figure 9: Stock detections for the victim: 5710, 5503, and aggregate 5551

```
[null@arch ~]$ for i in $(seq 1 10); do sshpass -p x ssh -o StrictHostKeyChecking=no fakeuser@172.21.0.5 exit; done;
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
Permission denied, please try again.
[null@arch ~]$ docker exec -it victim-ubuntu bash
root@victim-ubuntu:/# tail -5 /var/log/auth.log
Sep 24 10:39:07 victim-ubuntu sshd[6750]: Invalid user fakeuser from 172.21.0.1 port 54974
Sep 24 10:39:07 victim-ubuntu sshd[6750]: pam_unix(sshd:auth): check pass; user unknown
Sep 24 10:39:07 victim-ubuntu sshd[6750]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ru
ser= rhost=172.21.0.1
Sep 24 10:39:10 victim-ubuntu sshd[6750]: Failed password for invalid user fakeuser from 172.21.0.1 port 54974 ssh2
Sep 24 10:39:11 victim-ubuntu sshd[6750]: Connection closed by invalid user fakeuser 172.21.0.1 port 54974 [preauth]
root@victim-ubuntu:/# exit
exit
[null@arch ~]$ |
```

Figure 10: Repeat run against the victim IP for custom-rule validation

6 Detection 2: Custom Rule 100100

A per-attempt level-5 alert is noise; the SOC-relevant signal is the aggregate: 5 failures in 60 seconds at level 10, tagged T1110.001. I created the rule in the dashboard editor (Management → Rules → Files → `local_rules.xml`):

```
<group name="homelab,">
  <rule id="100100" level="10" frequency="5" timeframe="60">
    <if_matched_sid>5710</if_matched_sid>
    <description>SSH brute force, 5 fails in 60s</description>
    <mitre><id>T1110.001</id></mitre>
  </rule>
</group>
```

Two syntax failures preceded the working version, both caught by `wazuh-analysisd -t` (warning 7615, rule ignored): first a comma-plus-space list in `if_matched_sid`, then a spaceless list. `if_matched_sid` accepts exactly one rule ID, so the final rule watches 5710 only, which matches this scenario since every attempt targets a nonexistent user. After a manager restart and a fresh brute force, rule 100100 fired at level 10 (fired 4 times in the test window).



Figure 11: Custom rule 100100 in the dashboard ruleset editor

547 hits				
Sep 23, 2026 @ 14:17:06.250 - Sep 24, 2026 @ 14:17:06.250				
Export Formatted Reset view 637 available fields Columns Density 1 fields sorted Full screen				
timestamp	agent.name	rule.description	rule.level	rule.id
Sep 24, 2026 @ 14:16:46.0...	victim-ubuntu	SSH brute force, 5 fails in 60s	10	100100
Sep 24, 2026 @ 14:16:44.0...	victim-ubuntu	PAM: User login failed.	5	5503
Sep 24, 2026 @ 14:16:44.0...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 14:16:44.0...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 14:16:42.0...	victim-ubuntu	PAM: User login failed.	5	5503
Sep 24, 2026 @ 14:16:42.0...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 14:16:40.0...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 14:16:38.0...	victim-ubuntu	PAM: Multiple failed logins in a small period of time.	10	5551
Sep 24, 2026 @ 14:16:38.0...	victim-ubuntu	SSH brute force, 5 fails in 60s	10	100100
Sep 24, 2026 @ 14:16:38.0...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 14:16:36.1...	victim-ubuntu	PAM: User login failed.	5	5503
Sep 24, 2026 @ 14:16:36.0...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710
Sep 24, 2026 @ 14:16:36.0...	victim-ubuntu	sshd: Attempt to login using a non-existent user	5	5710

Figure 12: Custom rule 100100 fired at level 10 alongside stock rules

Document Details		View surrounding documents	View single document
Table	JSON		
t _index	wazuh-alerts-4.x-2026.09.24		
t agent.id	001		
t agent.ip	172.21.0.5		
t agent.name	victim-ubuntu		
t data.srcip	172.21.0.1		
t data.srcuser	fakeuser2		
t decoder.name	sshd		
t decoder.parent	sshd		
t full_log	Sep 24 11:16:45 victim-ubuntu sshd[7678]: Failed password for invalid user fakeuser2 from 172.21.0.1 port 57892 ssh2		
t id	1790248606.1646151		
t input.type	log		
t location	/var/log/auth.log		
t manager.name	wazuh.manager		
t predecoder.hostname	victim-ubuntu		
t predecoder.program_name	sshd		
t predecoder.timestamp	Sep 24 11:16:45		
t previous_output	Sep 24 11:16:43 victim-ubuntu sshd[7678]: Invalid user fakeuser2 from 172.21.0.1 port 57892 Sep 24 11:16:42 victim-ubuntu sshd[7676]: Failed password for invalid user fakeuser2 from 172.21.0.1 port 57884 ssh2 Sep 24 11:16:40 victim-ubuntu sshd[7676]: Invalid user fakeuser2 from 172.21.0.1 port 57884 Sep 24 11:16:39 victim-ubuntu sshd[7674]: Failed password for invalid user fakeuser2 from 172.21.0.1 port 57880 ssh2		
t rule.description	SSH brute force, 5 fails in 60s		
# rule.firedtimes	4		
# rule.frequency	5		
t rule.groups	homelab		

Figure 13: Alert detail: victim-ubuntu agent, attacker source IP, MITRE tag, frequency 5

6.1 Threshold rationale

A human mistypes roughly 1–3 passwords, then succeeds or stops; scripted guessing does dozens per minute. Five failures in 60 seconds catches automation fast in a quiet lab without flagging one fat-

fingering session. The stock sshd aggregate (5712) uses 8 in 120 seconds — stricter, tuned for busy servers. The 5/60 choice validates faster against a near-zero baseline. Rule source is committed as `detections/local_rules.xml`.

6.2 False positives

Known false positives: a legitimate user retrying five times in a minute (wrong keyboard layout, expired password) raises a level-10 alert on a human — mitigate with allowlists or by correlating with a subsequent successful login. Automation with a stale key (ansible, CI, monitoring logins) retrying on schedule — mitigate by allowlisting automation source IPs. Accepted scope flaw: the rule has no `same_source_ip`, so the count is global across agents; five users failing once each on five hosts would fire it. Acceptable with one victim, but production needs per-source scoping. There is also no `ignore` window, so a sustained brute force emits one alert per five events (four fired in the test run); add `ignore="60"` after tuning. Tuning loop: run one week, compare rule 100100 hits against confirmed incidents, then adjust frequency and timeframe.

7 MITRE Mapping

Rule	Event	Technique
5710	Invalid-user SSH login failure (stock)	T1110.001
5503	PAM authentication failure (stock)	T1110.001
5551	Multiple failed logins, small period (stock)	T1110.001
100100	5 SSH failures in 60s (custom)	T1110.001

8 Next Steps

Planned extensions: file integrity monitoring on `/etc` of the victim, an active-response block on brute-force sources, a second scenario (unauthorized `sudo` attempts), and a Windows agent for cross-platform coverage.